237%=120 or age%=0:

1%4

1=1

While Loop Task with Pen

Part A - Reverse Number

Question - 1

Reverse a given number.

Example : 153

output : 351

Step-1:

To reverse a number 153

-) First take out the last digit 153 %. 10 = 3

- Delete the last digit and get remaining part 153 // 10 = 15

- storing the last digit in other variable

x = 3

- Next we should multiply x with 10 and add it to next extracted number until none.

3 * 10 + 15 %. 10 = 30 + 5 = 35

35 * 10 + 1 %. 10 = 350 + 1 = 351

Step-2:

input - 153 , rev = 0

1) -> 153 %. 10 = 3 # extracted number

-> 153 // 10 = 15 # deleting extracted number

-> rev = 0 * 10 + 3 = 3 # our reverse number

2)

-> 15 %. 10 = 5

-> 15 // 10 = 1

-> rev = 3 * 10 + 5 = 35

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3) → 1 % 10 = 1

→ 1 // 10 = 0

→ rev = 35 * 10 + 1 = 351

Step-3

```
num = int(input())
rev = 0
while (n>0):
    n = num % 10
    num // = 10
    rev = rev * 10 + n
Print(rev)
```

o/p: 153
351

(2) Reverse a number and check it is a palindrome.

Ex: Input : 121
output : palindrome.

Step1: input - 121

→ Reverse the number first by

121 % 10 = 1
121 // 10 = 12 } 1st iteration
rev = 0 * 10 + 1 = 1

12 % 10 = 2
12 // 10 = 1 } 2nd iteration
rev = 1 * 10 + 2 = 12

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1 % 10 = 1

1 // 10 = 0

rev = 12 * 10 + 1 = 121

} 3rd & final iteration

→ Check it input == rev, if true then palindrome.

Step-2:

→ num % 10 # Extracting last digit

→ num // 10 # deleting

→ rev * 10 + num # deleteving our rev

→ if (num == rev) # checking the condn

Step-3:

Code:

```
num = int(input())
inp = int(input())
num = inp
rev = 0
while (num > 0):
    n = num % 10
    num // = 10
    rev = rev * 10 + n
if (inp == rev):
    print('Palindrome')
else:
    print('Not')
```

Output:

input - 121

output - palindrome.

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Q3: Reverse only the last N digits of number.

Example:

input = 123456

N = 3

output = 123654

Step-1:

input: 123456, n = 3

→ Reverse only upto n iteration.

1. 123456 % 10 = 6

123456 // 10 = 12345

0 * 10 + 6 = 6

2. 12345 % 10 = 5

12345 // 10 = 1234

6 * 10 + 5 = 65

3. 1234 % 10 = 4

1234 // 10 = 123

65 * 10 + 4 = 654

→ Add remaining part as it is 123654.

Step-2:

until 'n' number of iterations

→ n. num % 10

num // 10

rev * 10 + n

append the remaining part as it is

Step-3:

```
num = int(input())
x = int(input())
rev = 0
a = 1
while a <= x:
    n = num % 10
    num = num // 10
    rev = rev * 10 + n
    a += 1
print(num, rev, sep='')
```

output:

12345
3
12543.

Part B - Count Digits

Q4) Find total number of digits in a number.

Step-1:

- We can do it by dividing the number continuously until it becomes zero.
- Counting the number of iterations will give answer.

- 12345

→ 12345 // 10 = 1234 → c = 1

1234 // 10 = 123 → c = 2

123 // 10 = 12 → c = 3

12 // 10 = 1 → c = 4

1 // 10 = 0 → c = 5

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Step-2:

→ num // 10 until num becomes '0'.

→ number of iterations.

Step-3:

```
num = int(input())
if (num == 0)
    count = 1
else :
    count = 0
    while (num > 0 :
        num = num // 10
        count = count + 1
print(count)
```

output: 1234

4

Q5) Count the total even digits in a number.

Example: 824731

output: 3

Step-1:

- First extract last digit and check whether it is even or not.
- Do it respectively until the number 0.

824731 % 10 = 1 check 1 % 2 == 0 fa

824731 // 10 = 82473

823473 % 10 = 3

check 3 % 2 == 0 F

823473 // 10 = 82347

82347 % 10 = 7 check 7 % 2 == 0 F

82347 // 10 = 8234

8234 % 10 = 4 check 4 % 2 == 0, T, c = 1

8234 // 10 = 823

823 % 10 = 3

823 // 10 = 82

82 % 10 = 2 check 2 % 2 == 0, T, c = 2

82 // 10 = 8

8 % 10 = 8 check 8 % 2 == 0, T, c = 3

→ finally c = 3

step-3:

```
num = int(input())
count = 0
while(num > 0):
    dig = num % 10
    if(dig % 2 == 0):
        count = count + 1
    num = num // 10
print(count)
```

output 8 2 3 4 7 3

3

Q6) Count total odd digits

```
num = int(input())
Count = 0
while num > 0 :
    dig = num % 10
    if (dig % 2 != 0) :
        count += 1
    num //= 10
print(count)
```

output: 8 2 4 7 3)

3.

Q7) Part C - Digit Extraction

Find the first digit of a number.

EM1 12 3 4 5 6

To find first digit continuously divide by 10 until the number is less than 10.

12345 6 // 10 = 12345 & 10

12345 // 10 = 1234 & 10

1234 // 10 = 123 & 10

123 // 10 = 12 & 10

12 // 10 = 1 < 10 → This is our first character or number

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```
num = int(input())
while num >= 10:
    num = num // 10
print(num)
```

o/p: 2345

2

Q8) Find last digit of a number.

Just perform a modulo division on the number with 10.

```
num = int(input())
print(num % 10)
```

o/p: 123456

Q9) Find sum of all digits.

```
num = int(input())
s = 0
while num > 0:
    n = num % 10
    s = s + n
    num // = 10
print(s)
```

o/p: 12345

15

Q10) product of all digits.

```
num = int(input())
s = 1
while num > 0:
    n = num % 10
    s *= n
    num // = 10
Print(s)
```

o/p: 1234

24

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